Public

Short Reach Optical Interconnect

Ron Swartzentruber, AMD

Bobby Lu, Lightelligence

OCP SROI Workstream Chairs



OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

Public

The Challenge: Copper Is Running Out of Road

THE PROBLEM

AI training clusters are expanding beyond single racks into multi-rack Pods spanning 30 meters. At 200 Gb/s per lane, passive copper tops out at ~1–1.5 meters.

Next-gen Pods of 500–1,000+ accelerators require multi-rack connectivity — creating tens of thousands of optical lanes per rack. Conventional datacenter optics (DR/FR class) optimize for the wrong impairments.

≤ 30 m
SROI Reach Regime

500–1000+
Accelerators in next-gen multi-rack Pods

10,000s
Optical lanes per rack in future systems

12–18 mo
Accelerator generation cycle — window is narrow

A new reach-defined optical approach is required.

Scale-up interconnect is no longer secondary — it is the primary performance bottleneck for AI training.

Public

Defining SROI: A Distinct Optical Reach Regime

OPTICAL REACH TAXONOMY

In-Package OIO	XSR / VSR (OIF)	**SROI (this paper)**	DR class (IEEE)	FR class (IEEE)
< 1 cm	1–10 cm	1–30 m	500 m	2 km+

Traditional Datacenter Optics optimize for:

- ✗ Fiber attenuation over long distances
- ✗ Chromatic dispersion compensation
- ✗ Optical signal-to-noise ratio budgets
- ✗ Complex equalization and DSP

SROI optimizes for:

- ✓ **Optical lane density at package boundary**
- ✓ **Energy efficiency per bit (pJ/b)**
- ✓ **Cost per lane at rack-scale volumes**
- ✓ **Reliability for memory-semantic fabrics**

SROI is protocol-agnostic — serves UALink, ESUN, PCIe Gen7, CXL, and NVMe within the same physical layer.

Public

The AI Fabric: Two Distinct Network Domains

SCALE-UP

Interconnects within a Pod

Reach:	< 30 meters
RTT:	< 1 microsecond
BW / XPU:	~1,000 GB/s
Semantics:	Load / Store (I/O coherent)
Protocol:	UALink, ESUN, NVLink
Today's link:	1–3 m copper (passive)
Driver:	SROI adoption

Scale-up is the primary performance bottleneck

SCALE-OUT

Interconnecting multiple Pods

Reach:	< 2 km
RTT:	< 10 microseconds
BW / XPU:	~400 Gb/s
Semantics:	Message Passing (RDMA)
Protocol:	InfiniBand, Ultra Ethernet
Today's link:	Pluggable optics
Driver:	UEC / InfiniBand evolution

Scale-out: well-served by existing standards

Public

System-Level Requirements for SROI Adoption

POWER

Single digits pJ/b per link end

Goal: ≤ 5 pJ/b (optics only) · ≤ 10 pJ/b (SerDes + optics)

10 pJ/b is the copper-to-optics adoption crossover. CPO today: 4–5 pJ/b

COST

Trending toward copper in $/Gbit

Goal: Competitive with AEC on $/Gbit/m basis

Passive copper: ~$0.05/Gbit, Gap is ~10×. Cost includes cooling & power

RELIABILITY

≤ 2 FIT per link end · 0 link-flaps / 1M dev-hrs

Goal: Silicon-grade FIT (approaching passive copper)

Pluggable: 50–100 FIT. Operating at ≤40°C reduces FIT exponentially.

LATENCY

Equivalent to removing ≥ 1 retimer

Goal: No retimer in optical path

Each retimer: ~50 ns. 1m fiber: ~5 ns. Retimer elimination is the key lever

Full requirements consolidated in Appendix A of the white paper.

Public

Technology Options: A Two-Tier Deployment Model

Near-Term (Now)	Mid-Term (1–2 yrs)	Longer-Term (2–3+ yrs)
DEPLOY NOW	**TRANSITION TARGET**	**LONG-TERM GOAL**
Pluggable Optics	**Near-Package Optics**	**Co-Packaged Optics**
LPO / RTLR (FPLO)	*NPO (CPX MSA)*	*CPO (CPX MSA) / OCI MSA BiDi*
Power: 5-6 pJ/b (LPO) / 8-10 pJ/b (RLTR)	**Power:** ~5 pJ/b (linear)	**Power:** 4–5 pJ/b (PAM4) / 3 pJ/b (DWDM)
FIT: Improving w/ SiPho PIC	**FIT:** SiPho-enabled, improving	**FIT:** 5× better than pluggable
Bandwidth: 1.6T → 3.2T+ (400G)	**Bandwidth:** 6.4T today → 12.8T soon	**Bandwidth:** ASIC-generation roadmap
Repair: Front panel — field replaceable	**Repair:** Open-box — board removal	**Repair:** Factory repair — high cost
Best serviceability. Entry point for multi-rack Pod deployments.	*Best near-term balance of power, density, and reliability.*	*Lowest power, best signal integrity. Proprietary today; MSA maturing.*

Three governing trade-offs:

1. Power efficiency vs. serviceability
2. Integration depth vs. supply chain maturity
3. Bandwidth scalability vs. reliability track record

Public

Optical Connectivity Nomenclature

Location	Optics At Front Panel	Optics Behind the Front Panel			
	Front Pluggable Optics FPO	Optics Outside The Package NPO - Near Package Optics		Optics Inside The Package CPO - Co-packaged Optics	
Name	**FPO Optics Plugged at Front Panel**	**NPO-PCB [1] (or OBO) Optics on PCB**	**NPO-HDI [2] Optics on HDI**	**CPO Optics with Standard Packaging**	**CPO-AP[3] Optics with Advanced Packaging**
Attachment	Receptacle on front panel	Socketed or Soldered On PCB	Socketed or Soldered On HDI	Soldered (Standard Packaging)	Soldered (Advanced Packaging)
Rough Pitch for High-speed	~0.5 - 1.0 mm	PCB: 0.6-1.0 mm	HDI: 0.3-0.5 mm	C4 bumps / ~ 110 µm	µbumps / ~ 30-50 µm
Electrical I/F for High-speed	High-speed > 22 dB / 200G	High-speed Flex/Cable: 13-18dB/200G PCB: 16-22 dB / 200G	High-speed HDI: 13-18dB/200G	High-speed, & D2D (e.g. UCIe-S, BoW)	High-speed, & D2D (e.g. UCIe-A, BoW)
Characteristics	• May use integrated copper attach, copper/flex, or PCB traces	• May use integrated copper attach, copper/flex, or PCB traces	• HDI traces	• Lowest loss • High BW edge density • Integrated SoC	• Lowest loss • Highest BW edge density • Highly integrated SoC
Interoperability & Binding	Replaceable with interoperable optics at any time in life cycle, generally hot swappable	• Soldered: Interoperable but not replaceable • Socketed: Replaceable when removing blade	• Soldered: Interoperable but not replaceable • Socketed: Replaceable when removing blade	• Interoperable at chiplet level but not replaceable	• Interoperable at chiplet level but not replaceable
Standards	Standardized	Standards being developed		Some standards exist, others remain to be developed	

[1] NPO-PCB implementations are sometimes referred to as On-Board Optics (OBO)

[2] Some on-package implementations of NPO-HDI are sometimes referred to as "CPO" (CPO-HDI)

[3] A specific implementation of CPO-AP using a combination of an interposer and µbumps for optics has been referred to as Co-Integrated Optics (CIO)

Public

OCP's Role: Define Requirements, Enable Convergence

Without early convergence on requirements, fragmented protocol-specific solutions will prevent the volume aggregation needed to drive cost down and mature reliability.

1 Define Optical Reach Categories

Intra-rack and inter-rack scale-up reach classes, aligned with IEEE/OIF electricals. Feed directly into IEEE 802.3, OIF, and JEDEC standards development.

2 Establish Tiered Reliability Frameworks

Qualification tiers for in-package, intra-rack, and inter-rack SROI regimes. Leverage AEC standards for CPO; define new baseline for pluggable scale-up.

3 Define Interoperable Interface Objectives

Optical interface targets that preserve multi-vendor ecosystems. Enable CPX MSA and OCI-MSA alignment without mandating a specific photonic implementation.

4 Align Roadmaps with Emerging Protocols

Coordinate SROI readiness with UALink, ESUN, optical PCIe, and CXL-based memory fabrics. Optical readiness must not be the gating constraint.

AI infrastructure design cycles are compressing. The window for establishing this framework is narrow.

Public

Helpful Links:

- SROI Home page: https://www.opencompute.org/community/short-reach-optical-interconnect
- Whitepaper: https://www.opencompute.org/documents/sroi-white-paper-1-0-0-pdf/
- SROI OCP academy course: OCP Academy SROI course

Public

Backup



Public

Copper Channel Reach Limits at 200G and 400G

BACKUP

Topology	Spec	Max Cable	Notes
Passive copper (backplane)	IEEE 802.3dj	~1–1.5 m	PCB + cable + connector, no active components
NPC fly-over cable	IEEE 802.3dj study	1.4 m external	Near-Package Copper reduces PCB trace loss
CPC connectors	IEEE 802.3dj study	1.5 m or longer	Co-Package Copper eliminates PCB route loss
Active re-driver	Per-system	~2.0 m	Re-driver cheaper; less compensation than re-timer
Active re-timer	Per-system	~2.5–3.0 m	Full spec budget; adds ~50 ns latency per stage
Optical required	—	> 3 m	No copper path at 200 G/lane beyond ~3 m
400G PAM4 (CPC-CPC)	OIF study (projected)	~1 m	Technology improvements required; CPC likely only viable topology

Source: IEEE 802.3dj Task Force studies; OIF 448G signaling investigations. See §5 of white paper.

Public

Scale-Up vs. Scale-Out: Key Differences

BACKUP

Characteristic	Scale-Up Network	Scale-Out Network	Primary DC Network
Primary Application	XPU-to-XPU within a compute domain	Inter-cluster / inter-Pod	General connectivity
Typical Scale	256 XPUs, up to 1,000+ XPUs	Up to 10,000 nodes	100k–1M endpoints
Reach / Distance	< 30 meters	< 2 km	≥ 2 km
Round-Trip Time	< 1 microsecond	< 10 microseconds	> 100 microseconds
Bandwidth per XPU	~1,000 GB/s	~400 Gb/s	Shared ~100–200 Gb/s
Communication Model	Coherent I/O (Load/Store)	Message Passing (RDMA)	Network
Key Driver	Ultra-low latency, massive BW	High throughput, cost efficiency	Scalability, reliability
SROI Relevance	**PRIMARY driver**	**Served by existing standards**	**Out of scope**

Public

Scale-Up Switching: Electrical vs. Optical Circuit

BACKUP

	Electrical Circuit Switching	Optical Circuit Switching (OCS)
Link topology:	Two optical segments: XPU→Switch, Switch→XPU	Single end-to-end path: XPU→OCS→XPU
Latency:	Higher — O/E/O conversion at switch	Lower — no O/E/O; inherently low latency
Interface power:	Consumed at switch for both I/O interfaces	Zero at switch — optical pass-through only
Loss budget:	Each segment has separate optical budget	Combined: both segments + OCS insertion loss
Flexibility:	Packet-level switching; dynamic reconfiguration	Fiber-level connectivity; ms–µs reconfiguration
Maturity:	Available today — NVSwitch, scale-up ASICs	MEMS, LCoS, Piezoelectric, Robot-arm deployed; waveguide emerging

OCS adds insertion loss that must be budgeted. Zero I/O power at switch is a key advantage. See Table 4-8.

Public

Looking Forward: 400G/Lane Optical Transmission

BACKUP

PAM4: Preferred Optical Format

- ✓ Lower optical receiver power requirements
- ✓ Less sensitive to interferometric intensity noise
- ✓ Relaxed optical return loss vs. PAM6/PAM8
- ✓ InP and $LiNbO_3$ demonstrated at 400G PAM4
- ✓ Commercially prototyped at OFC and ECOC 2025

SiPho plasma dispersion bandwidth too low for 400G PAM4. Hybrid integration with InP or $LiNbO_3$ required.

Emerging Alternatives

Coherent-Lite

Higher DSP complexity for greater reach, improved link margin, and better OCS tolerance. 1.6T coherent-lite pluggables emerging for campus/intra-DC applications.

OCI-MSA (March 2026)

Open, silicon-centric optical PHY for AI scale-up. NRZ+WDM ("slow-and-wide") roadmap. Supports pluggable, on-board, and CPO. Bandwidth scaled through wavelengths per fiber, not faster lanes.

At 400G: copper favors PAM6 (lower Nyquist = longer reach); optics favor PAM4. This divergence will shape gearbox SerDes design. See §9 and §10.

Public

Requirements Summary: Appendix A Reference

BACKUP

Appendix A of the white paper consolidates all system-level requirements into a single reference for optics providers and standards contributors.

Table A-1	**Top-Level System Metrics**	Power, cost, FIT rate, link-flap rate, latency, RTT — threshold and goal values with rationale.
Table A-2	**Optical Physical Layer Parameters**	Reach, fiber type, operating temperature, modulation format, lane rate, retiming architecture.
Table A-3	**Density & Radix Requirements**	Interconnect radix, linear/areal BW density at ASIC edge, fiber count at faceplate, aggregate XPU BW.
Table A-4	**Reliability & Qualification**	Screening, qualification standards (Telcordia/AEC/JEDEC), diagnostics, redundancy, laser reliability.
Table A-5	**Module Placement Comparison**	PLO/LPO/RTLR vs NPO vs CPO vs SWO across power, FIT, BW scalability, repairability, and supply chain.

Full requirements also available as standalone Word document: SROI_Appendix_A.docx

Public

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

